

Rhizosphere hydrophobicity limits root water uptake after drying and subsequent rewetting

Mohsen Zarebanadkouki  · Mutez Ahmed ·
Clemens Hedwig · Pascal Benard · Stanley J. Kostka ·
Anders Kastner · Andrea Carminati

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Abstract

Background and Aims Recent experiments showed that rhizosphere of several plant species turns temporarily hydrophobic after severe drying and subsequent rewetting. Whether or not such hydrophobicity limits root water uptake is not known.

Methods A set of experiments was performed to test whether rhizosphere water repellency negatively affects root water uptake. To this end, a commercial surfactant was used as a rewetting agent to facilitate the wettability of the rhizosphere of lupins (*Lupinus albus*) in a sandy soil. Lupin plants were grown in rhizoboxes and were subjected to a severe drying cycle. Then half of the plants were irrigated with water and half with the surfactant solution. Time-series neutron radiography technique was used to monitor water redistribution in the rhizosphere during irrigation. In a second experimental set-up, lupins were grown in a sandy soil partitioned in

five vertical compartments separated by a 1-cm layer of coarse sand (acting as a capillary barrier). Water and surfactant were injected in different compartments and the rehydration of the root tissues beyond the irrigated compartments was monitored with neutron radiography for 2–3 h. Root rehydration rates were used to estimate the water fluxes across the root-soil interface.

Results The rhizosphere of lupin roots in sandy soil irrigated with water remained partly dry for at least 2–3 h, while it was rapidly rewetted when irrigated with surfactant. Water flow into the roots irrigated with surfactant solution was 6.5 times faster than into the roots irrigated with water.

Conclusions These results prove that water repellency of the rhizosphere of lupins in sandy soils limited the water fluxes into the roots and root rehydration during the first two to three hours after irrigation. This might not always be negative, because it can limit water losses from roots to dry soil and therefore avoid severe root dehydration.

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M. Zarebanadkouki (✉) · M. Ahmed · P. Benard ·
A. Carminati
Chair of Soil Physics, Bayreuth University, Bayreuth, Germany
e-mail: mohsen.zarebanadkouki@uni-bayreuth.de

C. Hedwig
Division of Soil Hydrology, Georg-August University of
Goettingen, 37077 Goettingen, Germany

S. J. Kostka
Aquatrols Corporation of America, Paulsboro, NJ, USA

A. Kastner
Paul Scherrer Institute, 5232 Villigen, Switzerland

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Introduction

The ability of plants to extract water from the soil is determined by the hydraulic conductivity of the roots and their neighboring soil, the *rhizosphere* (Nobel and Cui 1992). The rhizosphere is defined as the soil region close to the roots and affected by root activity. Our

understanding of the role of the rhizosphere on root water uptake has been accelerated with the development of imaging methods such as X-ray CT, neutron radiography and tomography and magnetic resonance imaging (Pohlmeier et al. 2008; Segal et al. 2008; Moradi et al. 2011). Neutron radiography was employed by Carminati et al. (2010) to monitor the dynamics of water content in the rhizosphere of lupines during drying and rewetting cycles. The authors found that the rhizosphere was wetter than the bulk soil during the drying period and it stayed temporarily dry after rewetting. This dynamic behavior of the rhizosphere was interpreted as the effect of root secreted mucilage. Mucilage is a gelatinous material secreted by the root tips (Hinsinger et al. 2009). Numerous functions have been attributed to mucilage, including solubilisation and entrapment of nutrients (Walker 2003), lubrication of the root surface for improving soil penetration (Iijima et al. 2004), maintenance of the contact between roots and soil particles (Bengough and McKenzie 1997), soil aggregation (Watt et al. 1994), facilitation of root water uptake during drying, and hydration of the root cap (Iijima et al. 2004).

McCully and Boyer (1997) showed that mucilage of maize is capable of adsorbing large volumes of water, possibly explaining the high rhizosphere water content during drying. However, mucilage contains a small fraction of amphiphilic components such as lipids, which, upon drying, can cause water repellency in the rhizosphere (Hallett et al. 2003; Moradi et al. 2012).

Ahmed et al. (2016) showed that mucilage collected from maize turns hydrophobic upon drying. The contact angle of glass slides covered with maize mucilage was greater than 80° at concentrations of 0.04 mg cm⁻² (milligram of dry mucilage per surface). Further experiments with *Lupinus albus*, *Maize*, *Vicia faba* and *Triticum aestivum* confirmed that mucilage increased the contact angle, but the effect is species dependent (Zickenrott et al. 2016). Kroener et al. (2015) and Benard et al. (2016, 2017) showed that the effect of mucilage on soil water repellency depends on soil texture, with coarse textured soil being more susceptible to water repellency. Soils mixed with mucilage turn water repellent when a sufficient fraction of the soil particle surface is covered with mucilage. Coarser soils have a lower specific surface (it scales with the particle radius to the power of minus one) and thus less mucilage is needed to cover to make the soil water repellent.

Despite the mechanisms of water repellency in the rhizosphere being partly clarified, their consequences on

root water uptake remain totally unknown. This knowledge gap is primarily due to the difficulty of measuring water fluxes across the root - soil interface.

In previous studies, we developed a method to trace water flow in soil and roots combining neutron radiography with the injection of deuterated water D₂O as a tracer (Zarebanadkouki et al. 2013). The method was used to measure local water uptake of root segments exposed to severe drying in comparison with uptake of roots that were kept in wet soil regions (Zarebanadkouki and Carminati 2014). The results showed that root water uptake was reduced by a factor of 4–8 in those regions that were allowed to dry and then irrigated. Although, these observations showed that rhizosphere hydrophobicity was accompanied with a temporarily reduction in root water uptake, it was not possible to exclude that the reduced uptake was caused by a decrease in root hydraulic conductivity or cavitation in the xylem of root (North and Nobel 1997).

The objective of this study was to test whether rhizosphere water repellency is a cause of reduced root water uptake after drying and wetting cycles. To this end, a commercial surfactant was used as a rewetting agent (control) to improve the rhizosphere wettability. The capability of this surfactant to rewet the hydrophobic rhizosphere was tested in lupins grown in a sandy soil. The plants were let dry soil to a very low water content. Then, some plants were irrigated with water and some with water treated with surfactant. Time-series neutron radiography was used to monitor water redistribution in the soil and in the rhizosphere during irrigation. After confirming the rhizosphere hydrophobicity of lupins in sandy soil and the capability of the selected surfactant to increase the rhizosphere wettability, a second experimental set-up was conducted to quantify the effect the rhizosphere water repellency on root rehydration rates and root water uptake – i.e. on the fluxes of water across the root-rhizosphere interface. Lupins were grown in a sandy soil, which was partitioned into five vertical compartments separated by a layer of coarse sand that acted as capillary barrier. As in the first experiment the soil was let dry. Then, one vertical compartment was irrigated with water and the symmetric compartment with surfactant. The rehydration of the roots inside and beyond the irrigated compartments was monitored using neutron radiography for a period of 2–3 h.

The hypothesis was that the selected surfactant will rapidly rewet the rhizosphere and facilitate the water fluxes across the root-rhizosphere interface compared

to the case of plants irrigated with water. The assumption was that differences in root water uptake of plants irrigated with water and surfactant can be mainly attributed to the change in hydraulic conductivity of the rhizosphere and not the root properties. To test this hypothesis, we conducted a third experiment in which the effect of surfactant on the permeability of root tissues was assessed using a root pressure method (Knipfer and Steudle 2008).

Materials and methods

Materials: Surfactant

Surfactants reduce the surface tension of water and the contact angle between water and hydrophobic surfaces. Surfactants increase water infiltrability in water repellent soils (Bially et al. 2005; Lehrs et al. 2011). In turf, surfactants improve soil moisture and turf quality. Recently, surfactants have been shown to increase water use efficiency in crops (Jafarian et al. 2015; Ahmadi et al. 2017). We used a commercially available surfactant called ACA 2045 (Aquatrols Corporation of America, Paulsboro, NJ) - a rewetting agent used in turf. The surfactant was used at a concentration of 1 g per liter of water. At this concentration the surface tension was 40.20 mN m^{-1} (measured by the sessile drop method, Drop Shape Analyzer DSA25S, Krüss GmbH). ACA 2045 is a non-ionic surfactant composed of a blend of alkyl polyglycoside surfactant and a block copolymer surfactant. The block copolymer surfactant consists of ethylene oxide and propylene oxide.

Experiment 1: Rhizosphere rewetting

The ability of the selected surfactant to facilitate the rhizosphere wettability was tested *in situ* by monitoring soil water content distributions around the roots of lupins grown in a sandy soil that was let dry and subsequently rewetted. Neutron radiography was used to monitor the water redistribution in soil and rhizosphere during irrigation.

Lupine plants were grown in aluminium rhizoboxes (size of $30 \times 15 \times 1 \text{ cm}$) filled with a sandy soil. The soil was collected from artificial catchment of Chicken Creek located near Cottbus, Germany. The soil (sieved to a particle size smaller than 2 mm) consisted of sand, silt, clay at a respective fraction of 92%, 5%, 3% and

contained no organic matter. The rhizoboxes were placed horizontally with the detachable top face left open to enable the filling of the rhizoboxes. The sandy soil was poured into each rhizobox through a 2-mm sieve in order to minimize soil layering. Then, the top face of the rhizoboxes was closed and the rhizoboxes were gently placed vertically. A layer of coarse sand was added on the soil surface to reduce evaporation from the soil surface.

Lupine seeds were germinated on moist filter paper for a period of 48 h. The seeds were sown, and the plants were grown under controlled condition with a photoperiod of 14 h, light intensity of $300 \mu\text{mol m}^{-2} \text{ s}^{-1}$, day/night temperature of $25^\circ\text{C}/19^\circ\text{C}$, and relative humidity of 60%. The plants were irrigated every fourth day by slowly immersing the bottom of the rhizoboxes in a nutrient solution. The rhizoboxes had holes at the bottom that allowed capillary rise and drainage. The plants were kept at a water content of 0.15 to $0.30 \text{ cm}^3 \text{ cm}^{-3}$ during the growth period. Thirty days after planting, the irrigation was ceased until the soil water content reached an average value of 0.02 – $0.04 \text{ cm}^3 \text{ cm}^{-3}$ (it took ca. 5 days). At this time the neutron radiography experiment started.

Five rhizoboxes were irrigated with water and five with the surfactant solution. Time-series neutron radiography was used to monitor the water and surfactant redistribution in soil and rhizosphere during irrigation. Irrigation was performed during night time (between 23:00 and 04:00) from the soil surface with a constant flow of 1 ml min^{-1} for a period of 2 h using a peristaltic pump (see Zarebanadkouki et al. (2016) for more details). After 2 h, for one plant the irrigation was switched from water to surfactant and the soil moisture distribution was continued to be monitored with neutron radiography. The idea was that after application of surfactant the rewetting rates of the rhizosphere increase.

Experiment 2: Water fluxes across a hydrophobic rhizosphere

To investigate the effect of rhizosphere hydrophobicity on root water uptake, three lupine plants (*Lupinus albus*) were grown in aluminium rhizoboxes (size of $28 \times 28 \times 1 \text{ cm}$) filled with the same sandy soil used in experiment 1. The rhizoboxes were laid horizontally with the detachable top face left open to enable the filling of the rhizoboxes. The inner space of the rhizoboxes was partitioned into five vertically oriented compartments

by placing four grids with a length of 20 cm and width and thickness of 1 cm. The sandy soil was poured into each compartment through a 2-mm sieve in order to minimize soil layering. The soil was wetted slowly and the grids were removed. The space left empty between the compartments was filled with coarse sand (grain diameter 1.2–1.7 mm) acting as capillary barriers and disconnecting the adjacent compartments in the upper 20 cm of the rhizoboxes. The top face of the rhizoboxes was closed and the rhizoboxes were turned vertically. The plants were grown as described in experiment 1. At 34 days after plantation, the irrigation was stopped and let the plants to dry down the soil to a volumetric water content of 0.02–0.04 cm³ cm⁻³. At this point two symmetric compartments of the rhizoboxes were simultaneously irrigated at night time (when transpiration was negligible) as follows: one compartment was irrigated with water and other symmetric (opposite) compartment with surfactant (concentration of 1 g per liter). Water and surfactant were added at a rate of 10 ml min⁻¹ for a period of 3–5 min (using a peristaltic pump). This flow rate corresponded to a flux lower than the saturated hydraulic conductivity of the soil, so that there was no water ponding and the soil was not saturated. Time-series neutron radiography was used to monitor water redistribution in soil and roots during irrigation. The objective of this experimental set-up was to monitor the redistribution of water and surfactant from the root segments located in the irrigated compartments towards the root segments located in the dry compartments beyond the capillary barriers.

Neutron radiography

Neutron radiography is an imaging technique that allows quantitative monitoring of water distribution in thin samples at high spatial and temporal resolution. The technique consists of passing a parallel neutron beam through the sample. The transmitted neutrons are captured in a 2D radiograph and carry information of the attenuation of neutrons through the sample. The transmission of the neutron beam through the sample is described by the Beer-Lambert law:

$$\frac{I}{I_0} = e^{-\sum_{i=1}^n (\mu_i d_i)} \quad (1)$$

where I is the intensity of the attenuated neutron beam (number of neutrons m⁻² s⁻¹), I_0 is the intensity of the

incident neutron beam (number of neutrons m⁻² s⁻¹), d_i (m) is the thickness of the i^{th} -material composing the sample, and μ_i (m⁻¹) is the macroscopic neutron attenuation coefficient describing the probability of neutron interactions with the i -material per unit of thickness.

Neutron radiography was carried out at the cold neutron imaging beam-line ICON at the Paul Scherrer Institut (PSI), Switzerland. We used a sCMOS camera with an array of 2160 × 2560 pixels, resulting in a field of view of 13.3 × 16 cm and pixel size of 0.062 mm. Nine radiographs with marginal overlap were needed to scan the rhizoboxes of 28 cm × 28 cm.

The relation between water content and neutron attenuation is calculated using the Beer-Lambert law

$$-\log \left(\frac{I(x, y, t) - dc}{I_{dry} - dc} \right) = \mu_{H_2O} d_{H_2O}(x, y, t) \quad (2)$$

where x and y are the coordinates of the field of view, t is time, $I(x, y, t)$ is the transmitted beam intensity, I_{dry} is the transmitted beam intensity through a rhizobox filled with dry soil, dc is the dark current (an image recorded by the camera when the beam was off), d_{H_2O} is the thickness of water and μ_{H_2O} is the attenuation coefficients of water ($\mu_{H_2O} = 3.65$ cm⁻¹), which was measured using control samples with known contents of H₂O. In Eq. (2) is assumed that soil density did not change during irrigation. In the pixels where there was no root, the average soil water content $\bar{\theta}(x, y, t)$ [cm cm⁻¹] was calculated as

$$\bar{\theta}(x, y, t) = \frac{d_{H_2O}(x, y, t)}{d_{tot}} \quad (3)$$

where $d_{tot} = 1$ cm is the thickness of the inner space of the rhizobox in the beam direction. The sharp contrast between roots and the surrounding soil facilitated the segmentation of the roots from the soil. Roots were segmented using the algorithm *Roottracker 2D* developed by Anders Kaestner and described in Menon et al. (2007). After root segmentation, three to five non-overlapping roots of each plant were selected for further analysis.

In the pixels containing a root, the neutron attenuation is average of the attenuation due to the root and the soil in front and behind it – i.e. the root occupies only a few millimetres of the rhizobox thickness in the beam direction. The actual contribution of the root in the neutron attenuation was calculated assuming that the water content in the soil in

front of and behind of the root is equal to that of the soil at the sides of the roots in the radiographs. The

change in volume of water in the root (water content of root), ΔV_{root} , was calculated as

$$\Delta V_{root} = \frac{\sum_{i=1}^{i=nx} \sum_{j=1}^{j=ny} \left(-\log \left(\frac{I(root, t)}{I(root, t=1)} \right) + \log \left(\frac{I(soil, t)}{I(soil, t=1)} \right) \frac{(d_{tot} - d_{root})}{d_{tot}} \right)}{\mu_{H_2O}} \lambda^2 \quad (4)$$

where i and j are the x and y coordinates that represent the position of a pixel in the neutron radiographs, $i=1$ and $j=1$ are the first root pixels in the x and y direction, and $i=nx$ and $j=ny$ are the last root pixels. $I(root, t)$ is the neutron attenuation of the pixel containing root and $I(soil, t)$ is the average neutron attenuation in the soil near the root, t refers to the time after irrigation when $t=1$ is the time before irrigation, d_{root} is the thickness of root, and λ is the side length of the pixels.

Experiment 3: Root hydraulic conductance

The root pressure probe method (Knipfer and Steudle 2008) was used to measure the hydraulic conductance of roots immersed in water and surfactant. The aim of this experiment was to test whether the used surfactant alters the permeability of the root tissues in the short time – i.e. during a period of two hours that corresponds to the observation time of experiments 1 and 2.

The root pressure probe consists of a pressure transducer and a capillary that connects an excised root system to the probe. Excised roots are connected to the probe, with the tightness of the connection being obtained with seals of silicone material (Xantopren L blue, Heraeus Kulzer GmbH, Germany). Half of the capillary (diameter of 600 μm) is filled with silicon oil and the other half of the capillary facing the root is filled with water. The meniscus between water and oil is used to measure water volume injected into and sucked from the root. The probe is equipped with a metal rod that can push or suck water into or out of the root.

Plants were grown as described in experiment 2. When the soil was at a water content of 0.08 to 0.10 $\text{cm}^3 \text{cm}^{-3}$, the detachable plate of the rhizobox was opened and the soil was carefully washed away from the roots. Then, unbranched root segments with a length of 4 to 9 cm were excised and connected to the root pressure probe. The root pressure probe was laid horizontally and the root was supported inside a glass container (10 $\times$ 30 cm) in which the root was kept

immersed in water. When root pressure reached a constant value (it took around 60 to 120 min), a series of pressure clamp experiments were carried out (3–5 clamps) to induce a flow of water into the root. Root pressure was increased in 30 to 100 kPa increments and held constant at each increment for 10–120 s by continually adjusting the metal rod inside the pressure chamber of the probe. The distance that the metal rod (meniscus) inside the capillary had moved was used to calculate the flow rate into the root [$\text{m}^3 \text{s}^{-1}$]. The slope of the flow rate plotted against the applied pressure difference was used to calculate the total conductance of the root, K_{root} [$\text{m}^3 \text{s}^{-1} \text{MPa}^{-1}$] (Knipfer and Steudle 2008). After performing 3–5 clamp experiments, the surrounding bathing solution (water) was replaced with a surfactant solution (at equal concentration as in experiment 2) and the total hydraulic conductance was measured again by performing 3–5 clamp experiments. The second series of conductance measurements were carried out in 90 min after immersion of the root segments in surfactant. Note that this experiment did not have the objective to mimic experiment 1 and 2, but rather to test the effect of surfactant on the root tissue. For this reason, we did not expose the roots to dry soil. This would have also been technically challenging. Indeed, to avoid cavitation in the xylem, the roots were first immersed in water and then cut. On the contrary, by cutting the roots in dry soils and directly connecting them to the root pressure would have likely induced cavitation in the xylem and would have complicated the interpretation of the root pressure probe data.

Results

Experiment 1: Rhizosphere rewetting

Neutron radiography of lupins irrigated with water after a severe drying revealed that the region around the roots stayed temporarily dry while the adjacent bulk soil was

quickly rewetted (Fig. 1). In contrast, when plants were irrigated with surfactant, both rhizosphere and bulk soil were uniformly and quickly rewetted. Figure 1a, b show the water content distribution in the soil at 30 min after irrigation with water and surfactant, respectively. In these images, the darker the image the wetter the soil. The dry region around the roots of the plant irrigated with water (Fig. 1a) was still remarkable 120 min after irrigation (Fig. 1c). The water content of the bulk soil irrigated with surfactant appeared slightly lower than the soil irrigated with water. This was caused by the lower surface tension and capillary forces of the surfactant, which reduces the water content at any negative water potential (for wettable soils, such as the sandy soil we used). To illustrate the effect of surfactant, for one plant after 120 min irrigation with water, the irrigation was switched from water to surfactant. The results of this plant showed that the rhizosphere which was still remarkably dry was rapidly rewetted with the surfactant solution (Fig. 1e). Figure 1e shows the difference in water content between the image taken after 120 min of irrigation with water and the one taken 20 min later after irrigation with the surfactant solution. The soil water content in the rhizosphere increased after irrigation with the surfactant (dark regions), while the change in water content of the bulk soil was smaller.

Experiment 2: Water fluxes across a hydrophobic rhizosphere

In the second experimental set-up, the rehydration rate of roots after irrigation with water and surfactant solution was interpreted as an indication of root water uptake. Figure 2 shows the neutron radiograph of one of the three rhizoboxes before irrigation. We first describe the experimental results from this rhizobox and then present the results from all three rhizoboxes.

Two compartments (C2 and C4 in Fig. 2) were simultaneously irrigated, one with water (C2) and the symmetric (opposite) one with surfactant (C4). The infiltration experiments were monitored using time-series neutron radiography (Fig. 2b–d). The exposure time of one image was 23 s. Since three radiographs at slightly overlapping positions were needed to scan the entire upper part of the sample, the resulting temporal resolution was ca. 70 s. In this experimental set-up our attention was concentrated on the swelling of the root segments located in the central compartment C3. In fact,

swelling and rehydration of these root segments is related to the water flow across the root-rhizosphere interface of the root segments in C2 and C4. Figure 3 shows a close-up of the ratio between the neutron radiographs at $t = 40, 80, 140$ min after starting the irrigation and the radiograph taken just before irrigation started ($t = 0$). This ratio indicates the changes in water contents, with the gray values being proportional to the changes in water content: i.e. the darker the image, the bigger the increase in water content after irrigation. These results showed that after irrigation, the roots irrigated with the surfactant solution rehydrated faster than the roots irrigated with water.

To quantitatively compare water and surfactant treatments, we calculated the changes in volume of water in the root segments within the irrigated compartments (C2 and C4) and beyond the barrier (C3). As the changes in volume of water in the root segments (flow rate) are affected by the length of root segments immersed in water in the irrigated compartments, the volumes of water in the roots beyond the barriers were normalized by the length of the root segments in the irrigated compartment (C2 and C4). The results for the roots of the plant shown in Fig. 2 are plotted in Fig. 4. The results of a total of 11 roots per treatment from 3 plants are shown in Fig. 5. Note that for quantifications we selected roots that had similar length and were not overlapping with each other in the central compartment (C3). We analyzed approximately 8 roots per plant (ca. 4 roots located in the compartment irrigated with water and ca. 4 roots in the compartment irrigated with surfactant). Figures 4 and 5 show that rehydration of the root segments (change in volume of water in the root segments) irrigated with surfactant was faster than that of root segments irrigated with water. In the case of roots irrigated with surfactant, the rehydration rates of root segments beyond the capillary barrier were initially fast (during first ca. 40 min after irrigation) and then decreased and approached constant values with time when root segments were fully rehydrated and water reached the taproot where it was not monitored any longer. Note that rehydration rate reached a constant value at varying times in each root segment depending on the rate of root water uptake and the length of root segment beyond the capillary barrier where the volumes are monitored. In contrast, for the case of roots irrigated with water, the rehydration of root segments beyond the capillary barrier was negligible during first ca. 15 min after irrigation and then slowly increased over time and did not reach a constant value in 160 min after irrigation.

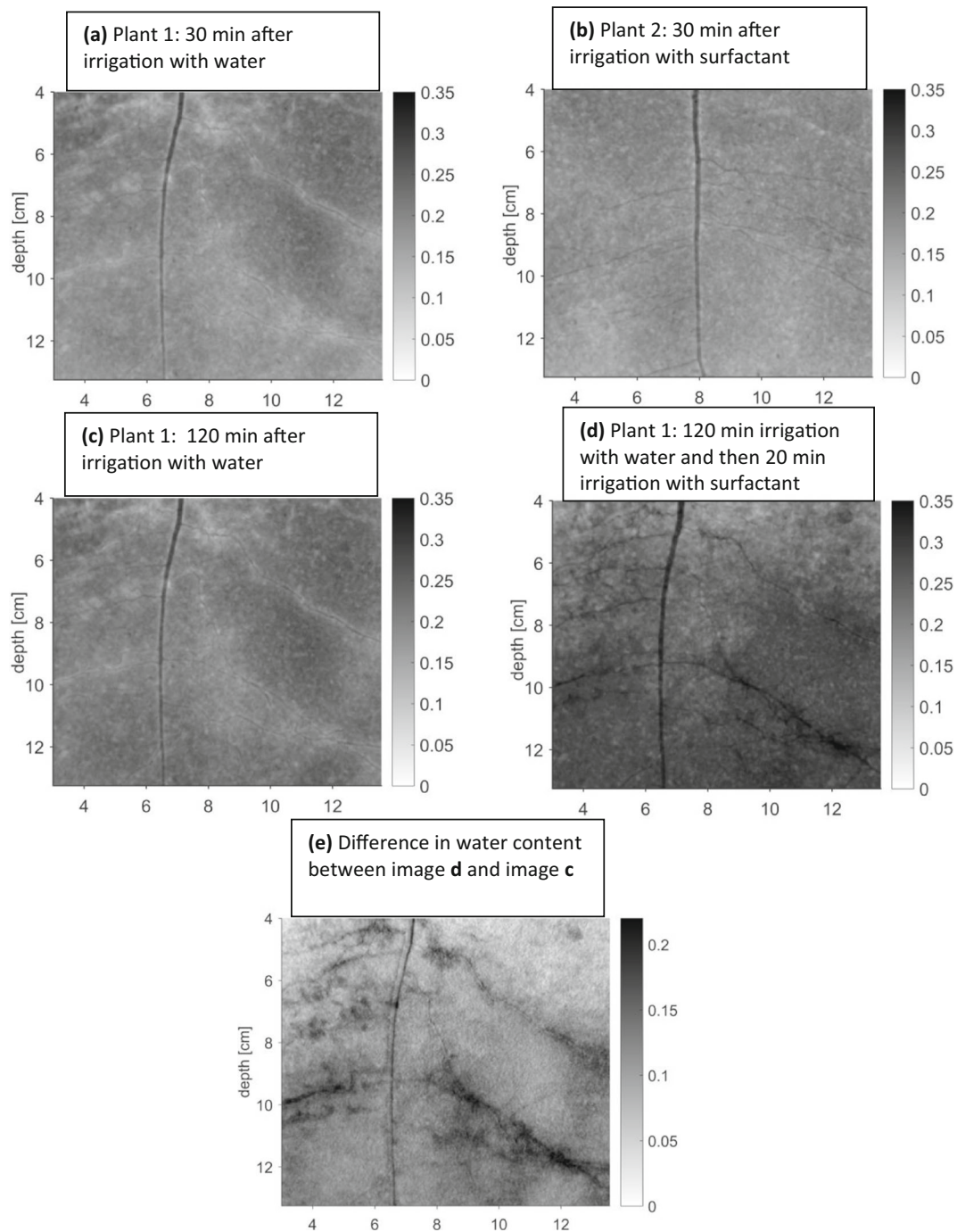


Fig. 1 Neutron radiography of one-month old lupines irrigated with water and surfactant. Images (a and b) show the water content distribution 30 min after starting the irrigation with water (plant 1) and surfactant (plant 2). Image (c) shows the water content distribution around the roots of plant 1 at 120 min after irrigation with water. Image 1d shows the water content distribution around the roots of plant 1 which was initially irrigated with water for 120 min (image (c)) and which was then irrigated with the surfactant; this image was taken 20 min after the irrigation with the surfactant. The

colour bars in images (a), (b), (c) and (d) show the water content in the rhizobox. Image 1e shows the change in soil water content of plant 1 120 min after irrigation with water and 20 min after irrigation with surfactant (i.e. it is the difference between image (d) and (c)). The colour bar shows the change in water content after irrigation with the surfactant. The results showed that the rhizosphere of lupin plants stayed temporarily dry after severe drying and subsequent rewetting with water while it was quickly and uniformly rewetted with surfactant

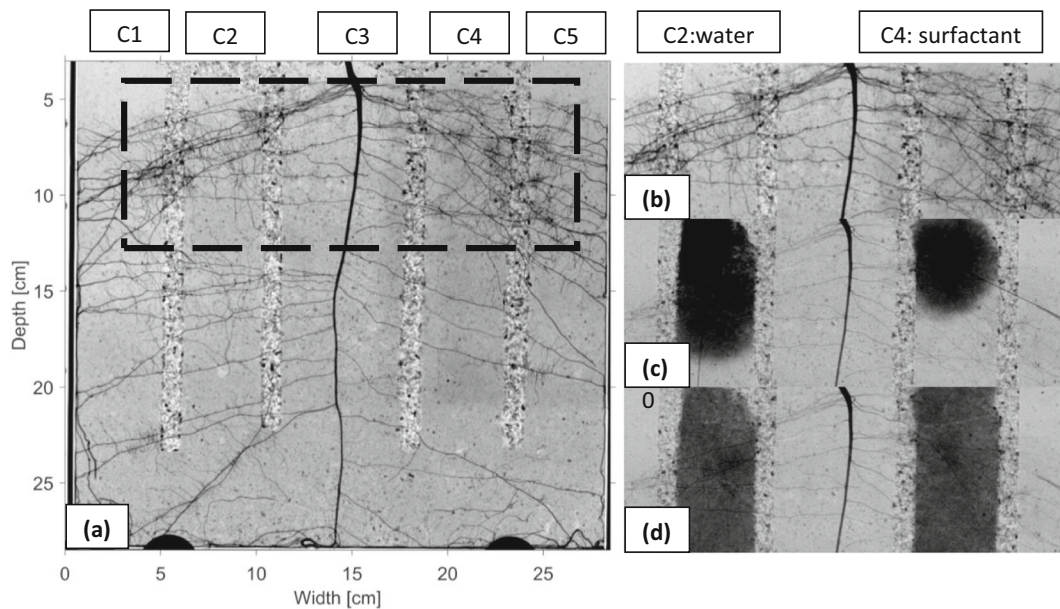


Fig. 2 **a** Neutron radiograph of a 40-day old lupine before the irrigation experiment started. In this image the darker is the image the wetter is the soil. The soil was partitioned into five compartments using a one cm layer of coarse sand acting as capillary barrier. Two symmetric compartments, C2 and C4, were irrigated with water and surfactant, respectively. Images **(b)**, **(c)** and **(d)** show neutron radiography of the upper part of the rhizoboxes (shown in image **a** with a dashed rectangle) before irrigation and around 3 and 30 min after starting the irrigation, respectively.

To relate the rehydration rate of the roots to the root water uptake (radial flux of water into roots), the following equation was used:

$$j_r 2\pi r L_{in} t = \Delta V_{in} + \Delta V_x + j_x \pi r^2 t \quad (5)$$

where the left side of Eq. (5) gives the radial water flow into the root segment immersed in water (with j_r being the radial flux of water [m s^{-1}], L_{in} the length of the root in the irrigated compartment [m] and r the root radius [m]), t is time [s], ΔV_{in} is the change in volume of water in the root segments immersed in water (or surfactant) [cm^3], ΔV_x is the change in volume of water in the root segment beyond the capillary barrier [cm^3], and $j_x \pi r^2$ is the axial flow of water towards the tap root and that does not contribute to root swelling (with j_x being the axial flux [m s^{-1}]).

Concerning the axial flow of water in saturated xylem, j_x , it was not measurable based on the rehydration rate of roots (root swelling) due to the restricted spatial resolution of neutron radiographs and the small cross section of xylem. However, as the experiments were performed at night, we may expect that its contribution

Image **(a)** is obtained from stitching 9 images with marginal overlaps and images **(b)**, **(c)** and **(d)** from stitching three images. The contrast of images has been adjusted to better see the roots in image **(a)** and **(b)**. Note that irrigation was performed simultaneously in the two compartments (C2 and C4). The wetting front seems to move faster in the compartment C2 of image **(c)**, but this is only because there is a delay of ca. 45–50 s between image C4 and C2

was small. The increase in volume of water in the root segments immersed in water, ΔV_{in} , was 5.0 ± 1.1 ($n = 11$) times faster in the plants irrigated with surfactant solution than with water. The volume of water in the root segment beyond the capillary barrier (central compartment, C3) increased 6.5 ± 0.7 ($n = 11$) times faster in plants irrigated with surfactant than with water. The differences between the treatments are a factor of 5 and 6.5 for ΔV_{in} and ΔV_x , respectively. Based on this consideration, we conclude that rhizosphere water repellency reduced the water fluxes into the root by at least 5 to 6.5 times.

Experiment 3: Root hydraulic conductance

Figure 6 shows the results of root pressure probe experiment and the hydraulic conductance of root segments immersed in water and surfactant. The total root conductance K_{root} [$\text{m}^3 \text{MPa}^{-1} \text{s}^{-1}$] was calculated from the slope of the linear regression of flow rate plotted against the applied pressure difference (Fig. 6b). The conductance of the root segments did not change after the roots

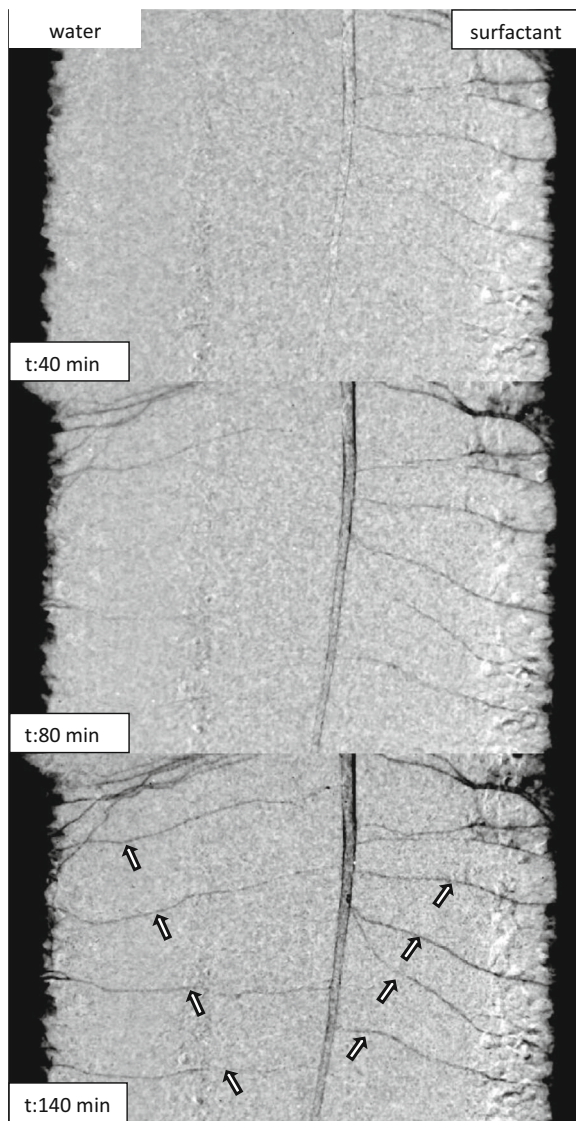


Fig. 3 Close-up of the rhizobox shown in Fig. 2, showing the difference between neutron radiographs at different times after irrigation and the one before irrigation (Fig. 2b). The grey values are proportional to the change in water content: the darker is the image the bigger is the change in water content. After irrigation, root segments beyond the capillary barrier (central compartment, C3) turned gradually dark, indicating that roots rehydrated because of water redistribution along roots. Root rehydration was faster in the compartment irrigated with surfactant. The arrows show the roots selected for further analysis

were immersed in surfactant solution ($p=0.1$). The average change (K_{root} of root irrigated with surfactant divide by K_{root} of roots irrigated with water) was 0.96 times with a standard division of 0.42). These measurements confirmed that within a short time after immersion in surfactants (90 to 120 min) the root permeability

was not significantly altered. We do not exclude that after longer exposure to surfactants root tissues might be damaged and the hydraulic conductance altered, but this is out of the scope of this paper.

Discussion

A surfactant was used as a rewetting agent to remove hydrophobicity of the rhizosphere of lupin plants that were grown in a sandy soil and that were subjected to a severe drying and subsequent rewetting cycle. The tested surfactant reduced water repellency of the rhizosphere and increased the rehydration rate of roots exposed to severe drying and subsequently rewetted. These findings prove that rhizosphere hydrophobicity temporarily limits the water-fluxes across the rhizosphere-root interface.

The first experiment with neutron radiography showing that the rhizosphere of lupins in sandy soil turns water repellent confirm former studies (Moradi et al. 2012). It was speculated that rhizosphere water repellency is caused by the hydrophobicity of dry mucilage (Moradi et al. 2012; Ahmed et al. 2016; Benard et al. 2017). Complementary experiments with seed and root mucilage support the fact the mucilage increases the contact angle of dry soils (Ahmed et al. 2016; Benard et al. 2017), however variations exist between seed and root mucilage, as well as among different plant species (Zickenrott et al. 2016; Naveed et al. 2017). Additionally, we do not exclude that microbial activity and repeated drying and rewetting events further increased water repellency in the rhizosphere.

The second experiment proves for the first time that rhizosphere water repellency limits root rehydration and root water uptake. Using an appositely designed experimental set-up, we found that the rehydration rate of roots irrigated with surfactant was 6.5 times faster than the roots irrigated with water. This finding supports the hypothesis of Zarebanadkouki and Carminati (2014), who measured a reduction in water uptake of roots that were first exposed to severe soil drying and then were irrigated. The authors found a reduction in water uptake of 4–8 times compared to roots of the same plant that were kept in a moist soil. However, in that study, the authors could not prove that the reduced water uptake was caused by the rhizosphere water repellency and not by a reduction of the root conductivity, for instance due to xylem

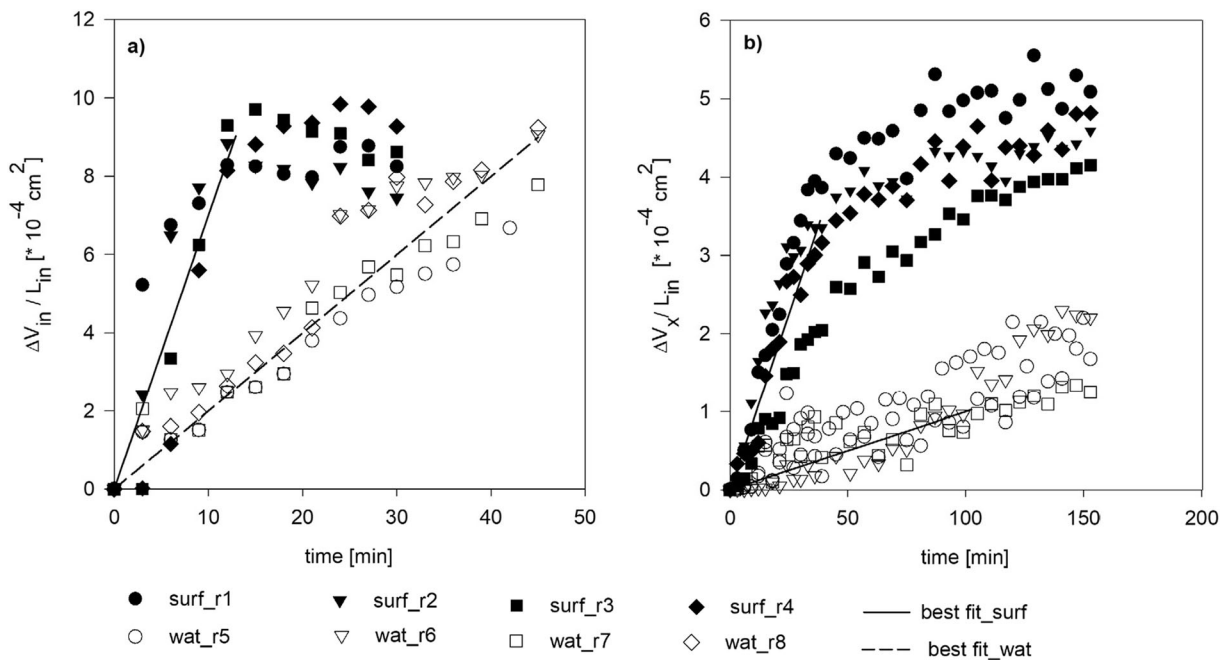


Fig. 4 **a** Change in volume of water in the root segments (ΔV_{in}) immersed in water (C2) and surfactant (C4). **b** Change in volume of water in the root segments beyond the capillary barrier in the central compartment of C3 (ΔV_x). The volumes of water are

normalized for the length of roots in the irrigated compartments (L_{in}). A linear function was fitted among different roots. The data refer to the roots indicated by arrows in Fig. 3

cavitation or to the loss of contact between roots and soil. In this study, it was demonstrated that rhizosphere hydrophobicity, which was removed by surfactant treatment, reduced the water fluxes across the root-

soil interface by 5–6.5 times, well explaining the observations of Zarebanadkouki and Carminati (2014).

It was experimentally challenging to show the effect of rhizosphere hydrophobicity on root water uptake. To

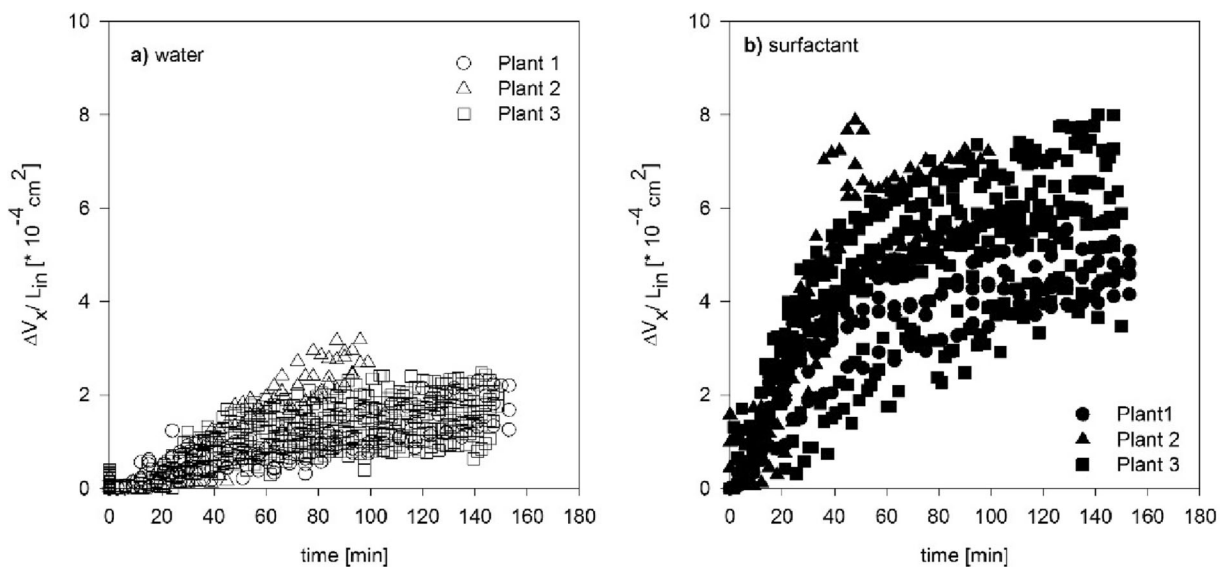


Fig. 5 Rehydration of the root segments in the dry compartment beyond the irrigated compartments with water (**a**) and surfactant (**b**). The data refer to three plants with a total of 11 roots per

treatment. Transport of water was faster in the roots irrigated with surfactant than with water

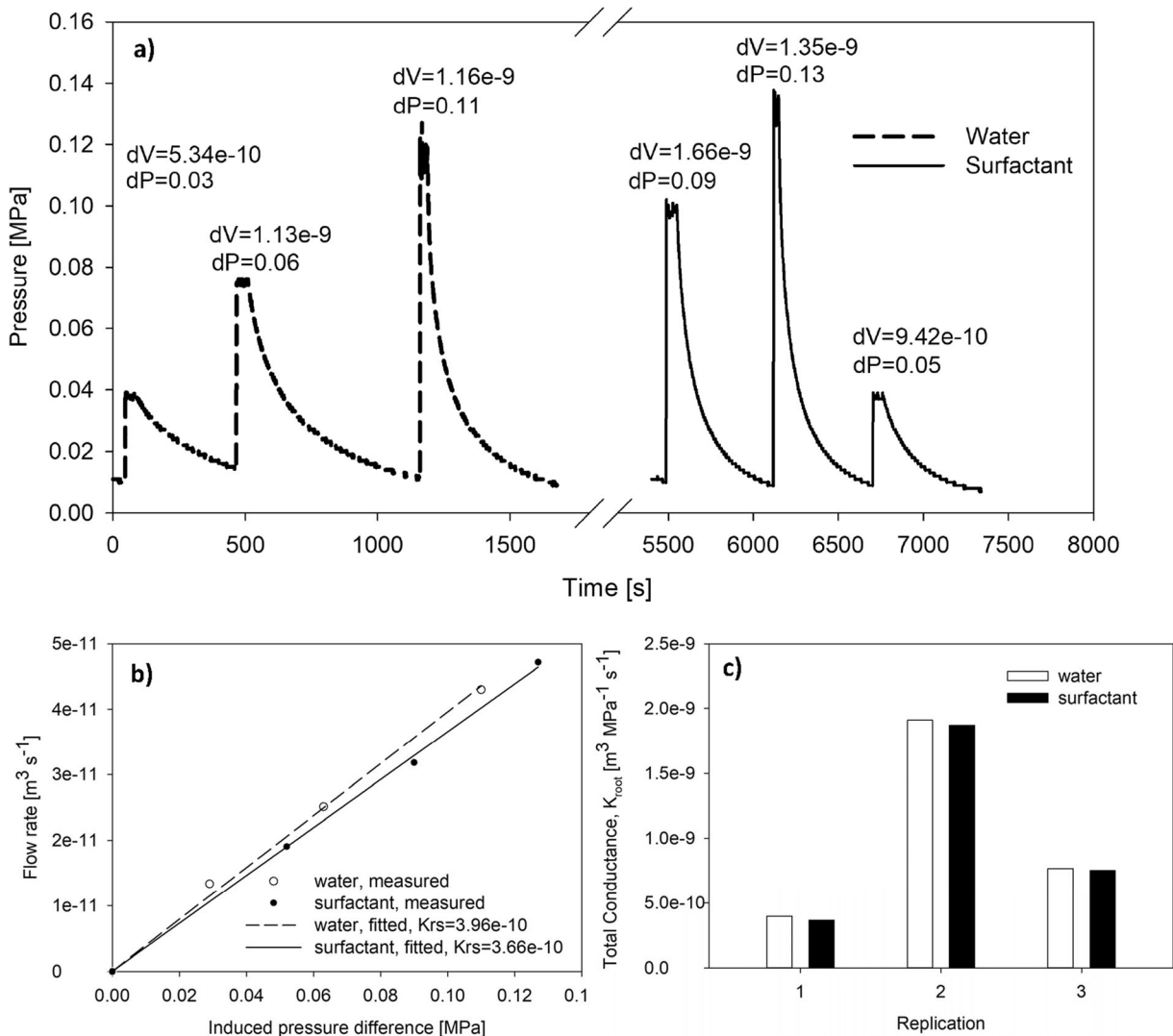


Fig. 6 **a** An exemplary root pressure probe experiment to determine the hydraulic conductance of a root segment immersed in water and then in surfactant. The differences in pressure (dP) and volume (dV) are given in MPa and m^3 respectively. **b** The relation between applied pressure difference and water flow. A linear function was fitted to the data of applied pressure difference and water flow and the slope of this line was taken as the root hydraulic

conductance [with a unite of $m^3 MPa^{-1} s^{-1}$]. **c** Hydraulic conductance of three roots which were first immersed in water and then immersed in surfactant solution for 90 min. Note that roots with different lengths were selected and therefore their hydraulic conductance was different - i.e. the longer the root the greater their hydraulic conductance

address this question, it was necessary to compare root water uptake across a hydrophobic and a non-hydrophobic rhizosphere while all the other factors such as root properties, xylem water potential and soil water potential are similar. The significance of this study is to use a surfactant as a rewetting agent of the hydrophobic rhizosphere and design an experimental set-up allowing us to compare root water uptake across hydrophobic (irrigated with water) and non-hydrophobic rhizosphere

(irrigated with surfactant solution on the same plant while all the parameters affecting root water uptake such as hydraulic properties of roots, xylem water potential and soil water potential were similar. However, a big concern was the effect of surfactant on the root hydraulic properties. This concern was addressed in experiment 3 in which we measured the hydraulic conductance of root segments immersed in water and surfactant. The results proved that the surfactant solution did not alter the

permeability of the roots during the time that root water uptake was measured in experiment 2 (two to three hours after irrigation). This experiment was performed to address short term effects of surfactant on root permeability and we cannot exclude the long-term effect of surfactants on plants.

Note that beside the rhizosphere, also the root epidermis and other regions within the root tissue might turn water repellent. For instance, xylem vessels can become temporarily hydrophobic upon drying (McCully et al. 2014). Therefore we cannot exclude that a potential hydrophobicity of the root epidermis might have further reduced the rehydration rates of the root tissues. Since separating the effects of rhizosphere and root conductances is challenging (if not impossible), it is probably more sensible to consider the conductance of the root-soil interface. In this sense, our study showed how the water repellency of the root-soil interface limited the root rehydration rates.

The findings of this study were based on experiments with lupines growing in sandy soil. To what extent can we generalize them? Concerning the soil, Kroener et al. (2015) and Benard et al. (2016) demonstrated that the effect of mucilage on soil water repellency depends on the soil texture: the coarser the soil, the lower the amount of mucilage needed to make it hydrophobic. Therefore, the rhizosphere of plants growing in coarse textured soils is more susceptible to water repellency. The results of this study are therefore representative of plants in coarse soils. Regarding the differences among plant species, Ahmed et al. (2016) recently reported similar degree of water repellency for maize. Zickenrott et al. (2016) measured the contact angle of mucilage from varying plants (*Lupinus albus*, *Vicia faba*, *Zea mays* and *Triticum aestivum*) showing high variability among plant species. Similar variability between chia seed mucilage and mucilage from young seedlings of barley and maize was recently reported by Naveed et al. (2017). Further studies with other plant species are therefore needed to support the broad generality of our study.

What are the implications of our study? We found that rhizosphere hydrophobicity delayed the rehydration rate of roots by 6.5 times. This effect lasted for at least 2–3 h, although it might last even longer. This might seem a short time and one might conclude that rhizosphere hydrophobicity has only a short time effect on water fluxes, with minor impact on plant growth and transpiration. Additionally, water repellency occurs just in severely dried soils and is probably limited to the

upper soil layers. However, there are cases when even a temporary reduction of the rhizosphere conductivity of the upper root segments might be critical. Consider for instance an irrigated field where plants are exposed to high transpiration demand (i.e. high temperature, solar radiation and low air humidity). In such a system, plants irrigated in the morning will be exposed to high transpiration rates within a few hours and the hydraulic conductivity of the rhizosphere might be a critical factor for sustaining high root water uptake rates. In a different environment, plants in arid and semi-arid regions where precipitation rates are short and rare would benefit from a prompt rehydration of their upper roots. In fact, in such scenarios, water might only infiltrate a few centimeters and roots should be fast in taking up this water before it is lost by evaporation. In this case rhizosphere water repellency might negatively impact plant growth. However, rhizosphere water repellency is likely to occur mainly (or only) in coarse textured soils, where water infiltrates deeper and might reach still hydrated roots with a less hydrophobic rhizosphere. A different interpretation is that although it temporarily limits root rehydration rates, rhizosphere hydrophobicity has the positive function of hydraulically disconnecting the root segments from the driest soil layers, therefore reducing leakage of water. In a larger scale simulation study in which plants with deep and shallow root architectures compete for water, it was shown that the rhizosphere water repellency might even be an advantageous trait. In fact, deep rooted plants benefit from making the upper soil water repellent therefore outcompeting shallow rooted plants which would then have less available water to grow (Zeppenfeld et al. 2017). In summary, the ecological and agriculture implications of the processes that we described here are highly dynamic and strongly dependent on soil and environmental conditions and vegetation type.

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